Srikar Molahalli

+91-9428852241 srikar.molahalli@gmail.com linkedin.com/in/srikar-molahalli github.com/Arknight007

Professional Summary

CS Major undergraduate at NMIMS MPSTME with experience in full stack JAVA development, cloud development, research driven software engineering and Rust based quantum computing . Shipped projects spanning in secure authentication systems, own programming language, solar scientific simulations as well as contributed as Cloud Computing Lead at Google Developer Group and Lead at Coding Club NMIMS, AIR 455 in GATE 2026, 94%ile JEE Mains, 12k JEE ADV, KVPY, NTSE Scholar

Education

- **NMIMS Deemed to be University** | Bachelor of Technology Tech Computer Science | 2.97/4 **July 2023 – Present**

Relevant Coursework

- Edge Cloud Computing
- Distributed System Design
- Operating Systems and Research
- Data Structures and Algorithm
- Computer Network and Security
- Database Management System
- Computer Architecture
- AIML: Deep Learning
- Quantum Computing

Experience

RTX Corporation Inc. Collins Aerospace **May 2025 – July 2025**
Junior Software Intern, Mission Systems Summer Engineering Internship Program *Bangalore, India*

- Improved the processing pipeline with CUDA along with multi threading, reducing the overall compute time by 20%.
- Automated parsing workflows using Python asyncio, shrinking turnaround from 8 hours to 2 hours, a 75% reduction.
- Expanded the integration test suite's multiprocessing by lifting code coverage to 95% and speeding releases by 40%.
- Setting up the Jenkins-driven CI/CD pipelines to build automatizations and trim manual integration work by 50%.

Indian Institute of Science Education and Research (IISER), Kolkata **May 2024 – July 2024**
Junior Summer Associate Research Intern *Kolkata, India*

- Studied instability patterns to support fuel-efficient mission planning for soft landings and yielding a 15% gain overall.
- Explored the spacecraft paths through simulation workflows and lowering landing deviation by 10% during evaluation.
- Modeled spacecraft motion with numerical analysis thus improving touchdown precision by 5% across key scenarios.
- Built an interactive dashboard for trajectory and flow insights with cutting interpretation time by 30% significantly.

Projects

Quasar Programming Language Compiler | *C, LLVM, Flex, Bison, x86 Assembly* | GitHub **October 2025**

- a custom statically-typed programming language, translating high-level source code into executable 32-bit x86 assembly.
- Implemented the lexical analysis and parsing, constructing a syntax tree for semantic validation and code generation
- Leveraged LLVM infrastructure for an optimized intermediate representation and then improving execution efficiency.
- Compiler phases including syntax semantic and final code emission then by processing 1,000+ lines of source efficiently.
- Achieved a 95% test coverage with a robust validation pipelines, ensuring correctness across diverse input programs.

Solar Positioning Equinox Simulation | *Java, JDBC/ODBC, JavaFX, JSwing, Eclipse* | GitHub **March 2025**

- Developed a Java-based interactive simulation that visualizes the Sun's position relative to Earth across all equinoxes
- Modeled four major astronomical events with smooth real-time animations, thus improving user engagement by 60%.
- Refined the interface for 30% lower latency, improving responsiveness and supporting a cleaner modular maintenance.
- Applied trigonometric computations to represent all equinox and solstice transitions with accurate real-time behavior.
- Designed a dynamic side-panel interface with 10+ inputs for dimension control, increasing simulation flexibility by 40%.

Borderless Financial Inclusion | *Tailwind CSS, TypeScript, React, Flask, Ethers.js, Web3* | GitHub **April 2025**

- Developed a Flask-backed DeFi platform across 5+ blockchains where by reducing cross-border remittance fees by 90%.
- Engineered a responsive frontend with React, three.js and Web3 libraries, improving UI rendering performance by 30%.
- Created a user specific, customized yield wallets with custom investment strategies, delivering up to 40% APY returns.
- Applied quantitative analytics principles across 7+ account to design scalable data models handling 10,000+ records.
- Integrated real-time blockchain transaction tracking with Ethers.js, enhancing transparency and user trust by 35%.

Skills & Achievements

Languages & Databases: Python, Java, C, C++, JavaScript, TypeScript, SQL, R, Julia, Rust, HTML, Tailwind CSS, Scala, Modern Angular, Angular,js Neo4j, PostgreSQL, MongoDB, BigQuery, G-APPSheets, Supabase, NEON, Snowflake.
Frameworks & Tools: React, Next.js, Node.js, Express.js, FastAPI, Flask, JavaFX, RESTfulAPI, Docker, Kubernetes, Azure DevOps, Google Cloud CI/CD, Jenkins, PyTorch, Ethers.js, Git, GitLab, Postman, Selenium, Qiskit, Ollama, Cirq
Cloud & DevOps: Microsoft Azure, Amazon Web Services, Google Cloud, Containerization, CI/CD, Monitoring, Linux
Leadership: Coding Club: Lead, Google Developer Group on Campus: Cloud Computing Lead, AIR 455 GATE 2026

Certifications & Publications

- **Certifications**: Google Cloud Computing Foundations Certificate, Networking & Security in Google Cloud, Claude with Amazon Bedrock, Claude with Google Cloud's Vertex AI, Harvard CS50x, CS50P, CS50AI, AWS AIF-C01, DeepLearning.AI
- **Publication**: *Beyond Relational: Leveraging Graph Databases for Efficient Real-time Data Processing*, Annacis Research and Event Centre, Vancouver Canada *to be published in – Inspire Intelligence Journal (ISSN: 3110-942X)*